CreditPath AI

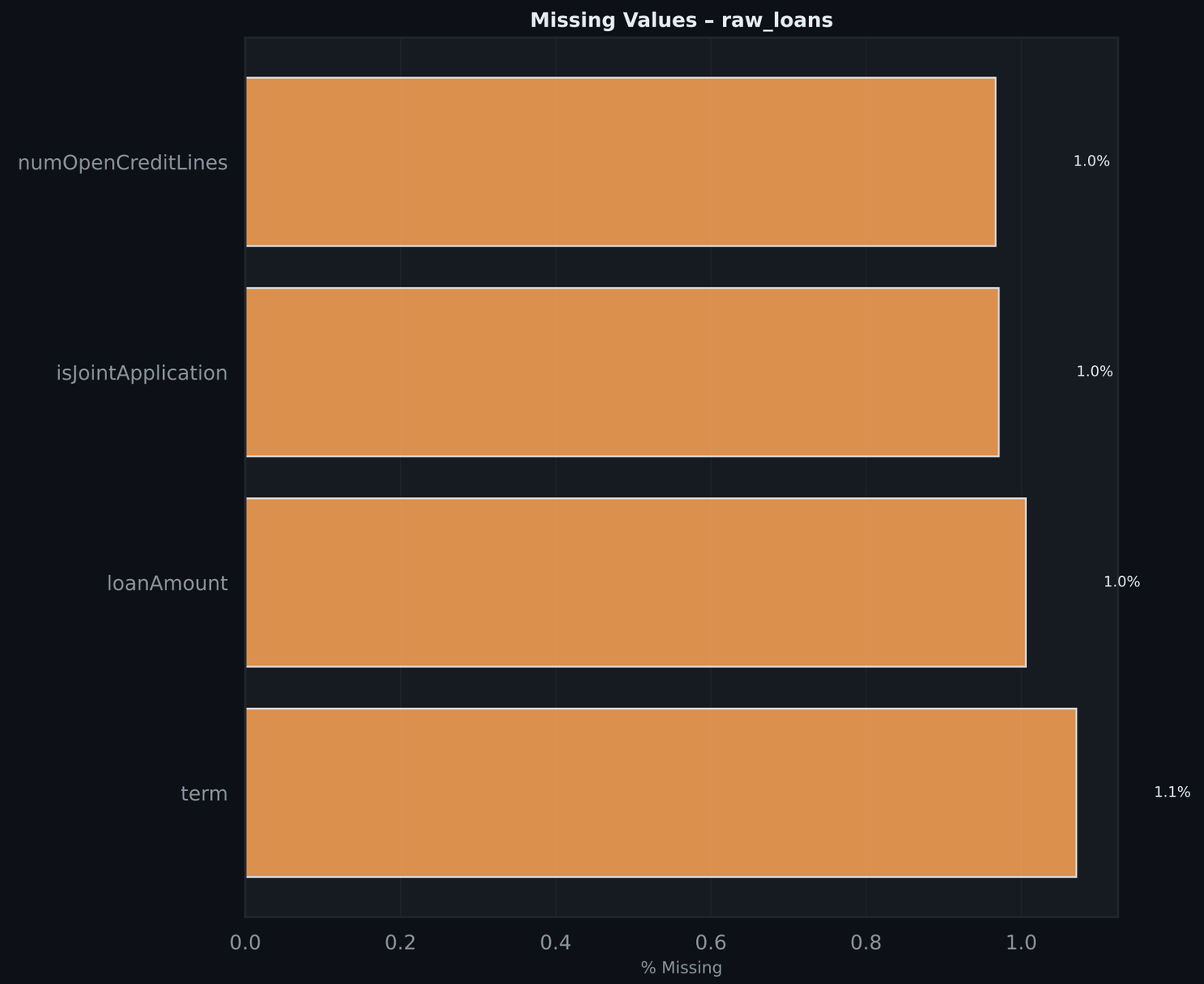
Exploratory Data Analysis Report

Dataset: processed_loans | Rows: 100,000 | Features: 43

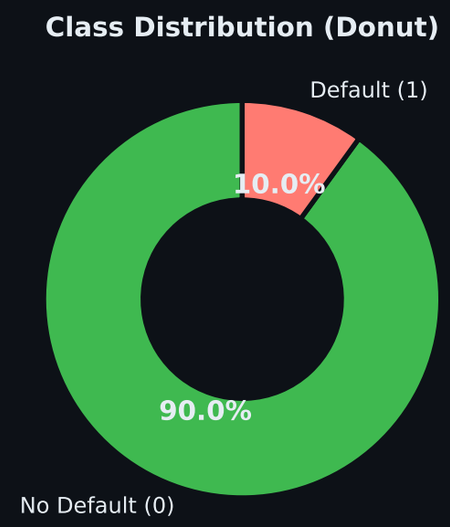
Target → loanStatus (0 = No Default | 1 = Default)

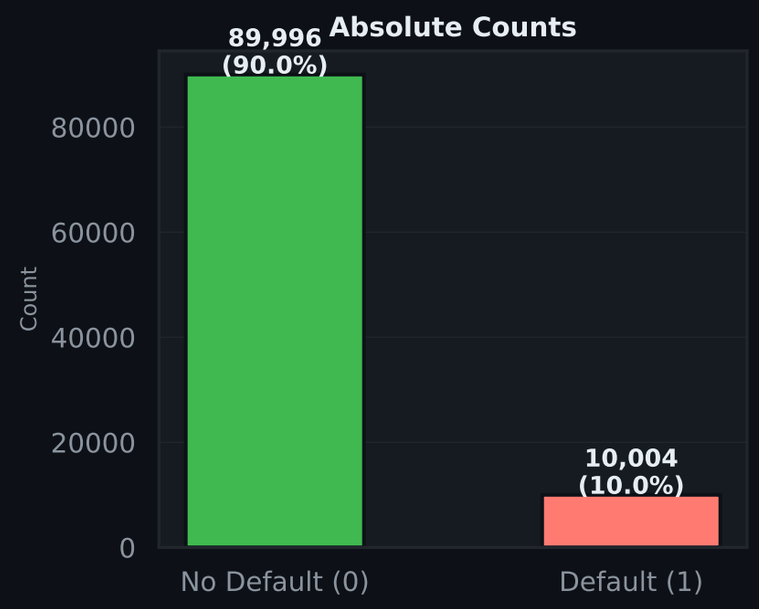
Dataset Overview

No Missing Values
in processed_loans ✓



Target Variable Analysis – loanStatus (0=No Default, 1=Default)



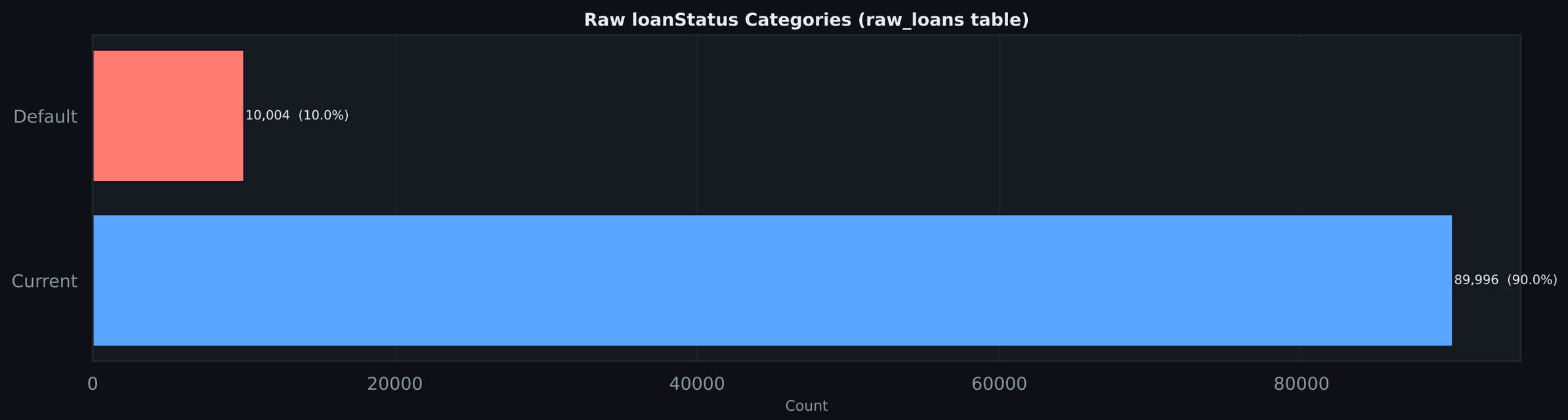


Key Statistics

```
Default Rate:       10.00%
No-Default Rate:    90.00%

Total Records:      100,000
Defaults:           10,004
Non-Defaults:       89,996

Imbalance Ratio:    1 : 9.0
```



Descriptive Statistics (processed_loans – numeric features)

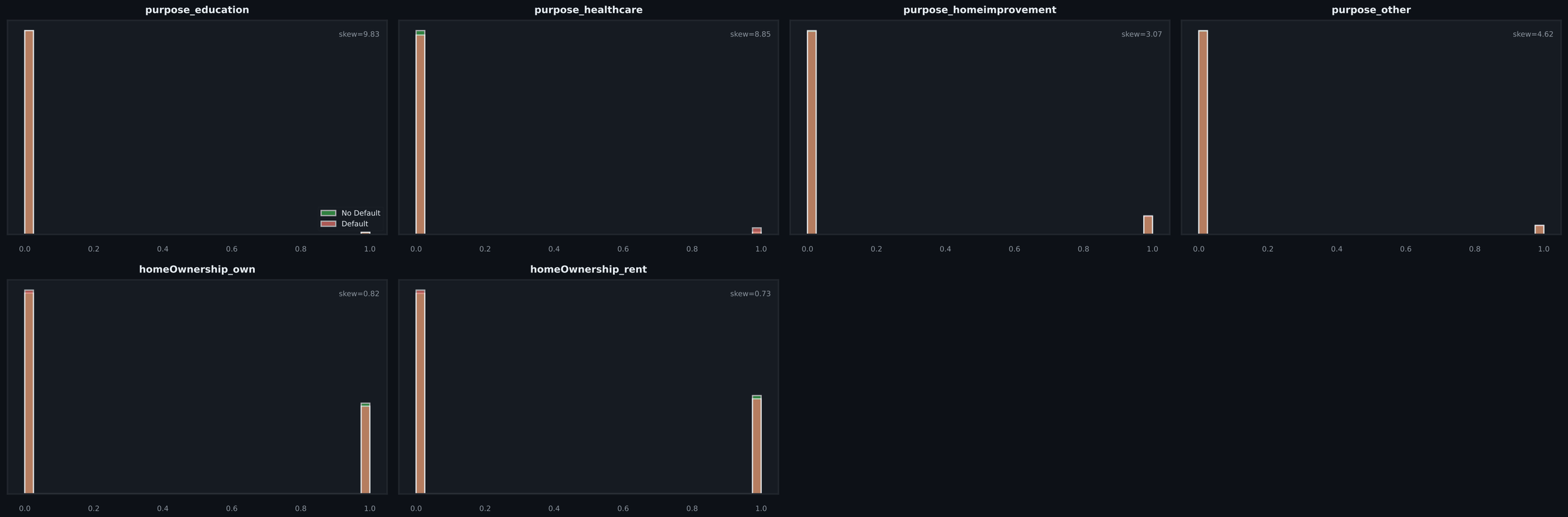
	count	mean	std	min	25%	50%	75%	max	skewness	kurtosis
isJointApplication	1e+05	0.0502	0.217	0	0	0	0	1	4.14	15.2
loanAmount	1e+05	2.07e+04	4.67e+03	8.02e+03	1.75e+04	2.07e+04	2.38e+04	3.32e+04	0.0586	-0.0949
interestRate	1e+05	11	3.82	4.02	7.98	10.5	13.4	21.6	0.566	-0.2
monthlyPayment	1e+05	552	162	102	434	534	655	986	0.433	-0.127
yearsEmployment	1e+05	4.14	3.49	0.5	1	4.14	4.14	10	0.699	-0.859
annualIncome	1e+05	5.47e+04	3.14e+03	4.63e+04	5.26e+04	5.48e+04	5.69e+04	6.32e+04	-0.0926	-0.0981
incomeVerified	1e+05	0.686	0.464	0	0	1	1	1	-0.802	-1.36
dtiRatio	1e+05	17.3	5.07	3.47	13.8	17.2	20.7	31.1	0.0191	-0.172
lengthCreditHistory	1e+05	10.5	9.76	1	4	6	15	31.5	1.14	-0.108
numTotalCreditLines	1e+05	15	4	3	12	15	18	27	-0.00084	-0.0648
numOpenCreditLines	1e+05	11.3	3.71	1.5	9	11	14	21.5	0.368	-0.128
numOpenCreditLines1Year	1e+05	7.59	3.36	1	5	7	10	17.5	0.685	0.11
revolvingBalance	1e+05	1.48e+04	2.49e+03	8.07e+03	1.32e+04	1.49e+04	1.65e+04	2.16e+04	0.017	-0.176
revolvingUtilizationRate	1e+05	65.1	15.2	22.9	54.5	65.2	75.6	100	-0.0215	-0.298
numDerogatoryRec	1e+05	0	0	0	0	0	0	0	0	0
numDelinquency2Years	1e+05	0.502	0.927	0	0	0	1	2.5	1.5	0.477
numChargeoff1year	1e+05	0.483	0.9	0	0	0	1	2.5	1.55	0.695
numInquiries6Mon	1e+05	0.724	0.898	0	0	0	1	2.5	0.86	-0.678
term_months	1e+05	48.2	9.78	36	36	48	60	60	-0.0281	-1.49
grade_score	1e+05	2.64	1.34	1.1	1.3	2.3	3.3	5.3	0.535	-0.888
loan_to_income_ratio	1e+05	0.379	0.0892	0.137	0.318	0.377	0.437	0.617	0.142	-0.0645
payment_to_income_ratio	1e+05	0.0101	0.00306	0.00169	0.0079	0.00977	0.012	0.0182	0.471	-0.0918
repayment_velocity	1e+05	0.0267	0.0049	0.0184	0.0228	0.0257	0.0313	0.043	0.286	-1.05
loan_amortization_rate	1e+05	437	134	61.6	340	423	520	800	0.507	-0.0931
open_credit_ratio	1e+05	0.7	0.137	0.333	0.583	0.7	0.812	0.967	0.0298	-1.1
recent_credit_velocity	1e+05	0.614	0.176	0.2	0.467	0.615	0.75	0.963	-0.0113	-1.04
inquiry_intensity	1e+05	0.112	0.162	0	0	0	0.182	0.455	1.22	-0.048
delinquency_density	1e+05	0.017	0.0294	0	0	0	0.0278	0.0694	1.19	-0.557
derogatory_density	1e+05	0	0	0	0	0	0	0	0	0
estimated_credit_limit	1e+05	2.39e+04	6.78e+03	8.17e+03	1.89e+04	2.27e+04	2.76e+04	4.06e+04	0.745	0.0626
credit_utilization_recomputed	1e+05	0.651	0.152	0.229	0.545	0.652	0.756	1	-0.0215	-0.298
log_loanAmount	1e+05	9.91	0.233	9.31	9.77	9.93	10.1	10.5	-0.436	0.00427
log_annualIncome	1e+05	10.9	0.0576	10.8	10.9	10.9	10.9	11.1	-0.218	-0.0838
log_revolvingBalance	1e+05	9.59	0.171	9.14	9.48	9.61	9.71	10.1	-0.358	-0.0987
purpose_business	1e+05	0.0321	0.176	0	0	0	0	1	5.31	26.2
purpose_debtconsolidation	1e+05	0.813	0.39	0	1	1	1	1	-1.61	0.59
purpose_education	1e+05	0.01	0.0996	0	0	0	0	1	9.83	94.7
purpose_healthcare	1e+05	0.0123	0.11	0	0	0	0	1	8.85	76.4
purpose_homeimprovement	1e+05	0.0812	0.273	0	0	0	0	1	3.07	7.41
purpose_other	1e+05	0.0412	0.199	0	0	0	0	1	4.62	19.3
homeOwnership_own	1e+05	0.31	0.463	0	0	0	1	1	0.821	-1.33
homeOwnership_rent	1e+05	0.328	0.469	0	0	0	1	1	0.733	-1.46
loanStatus	1e+05	0.1	0.3	0	0	0	0	1	nan	nan

Feature Distributions (Page 1/4)

Feature Distributions (Page 2/4)

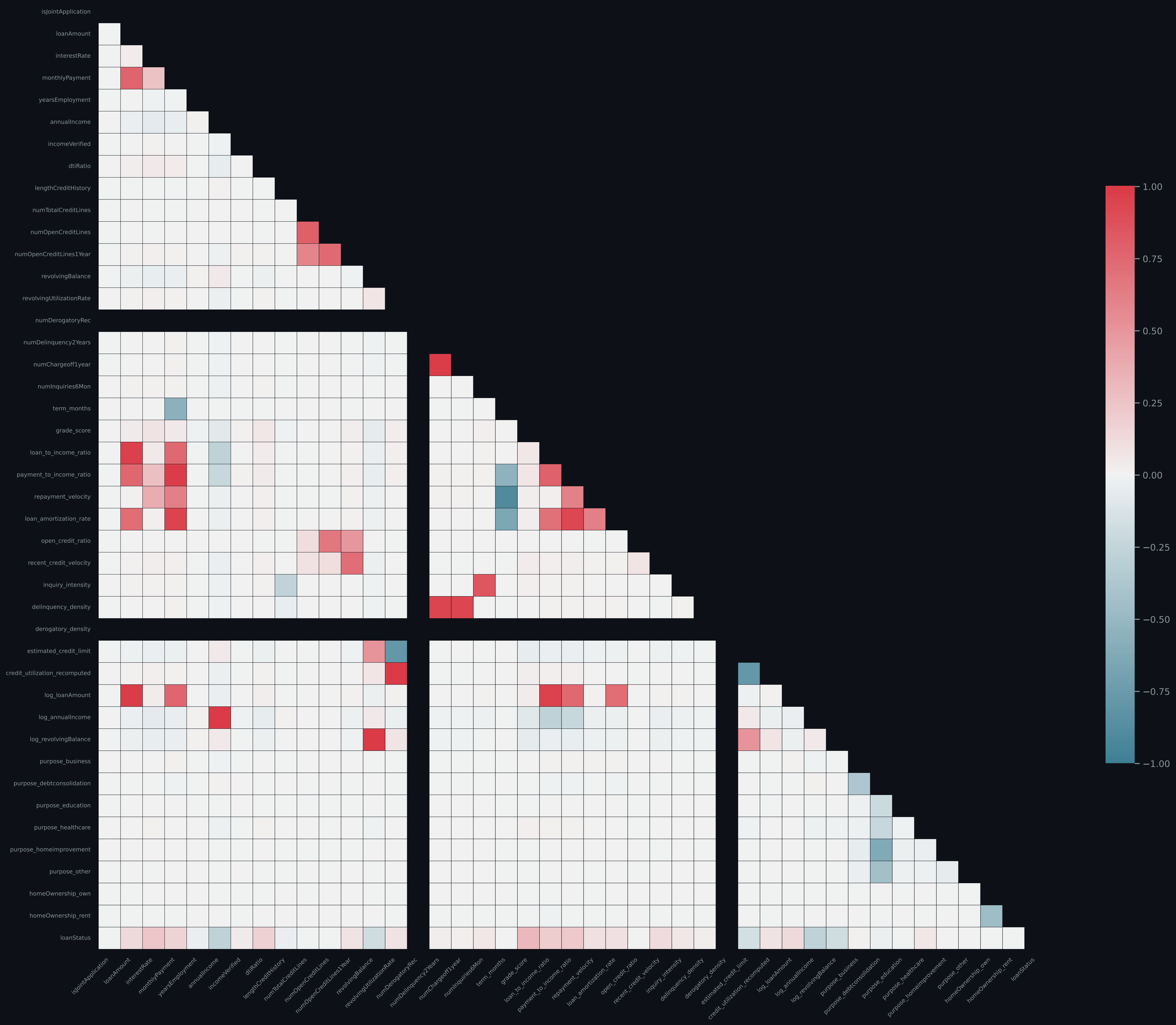
Feature Distributions (Page 3/4)

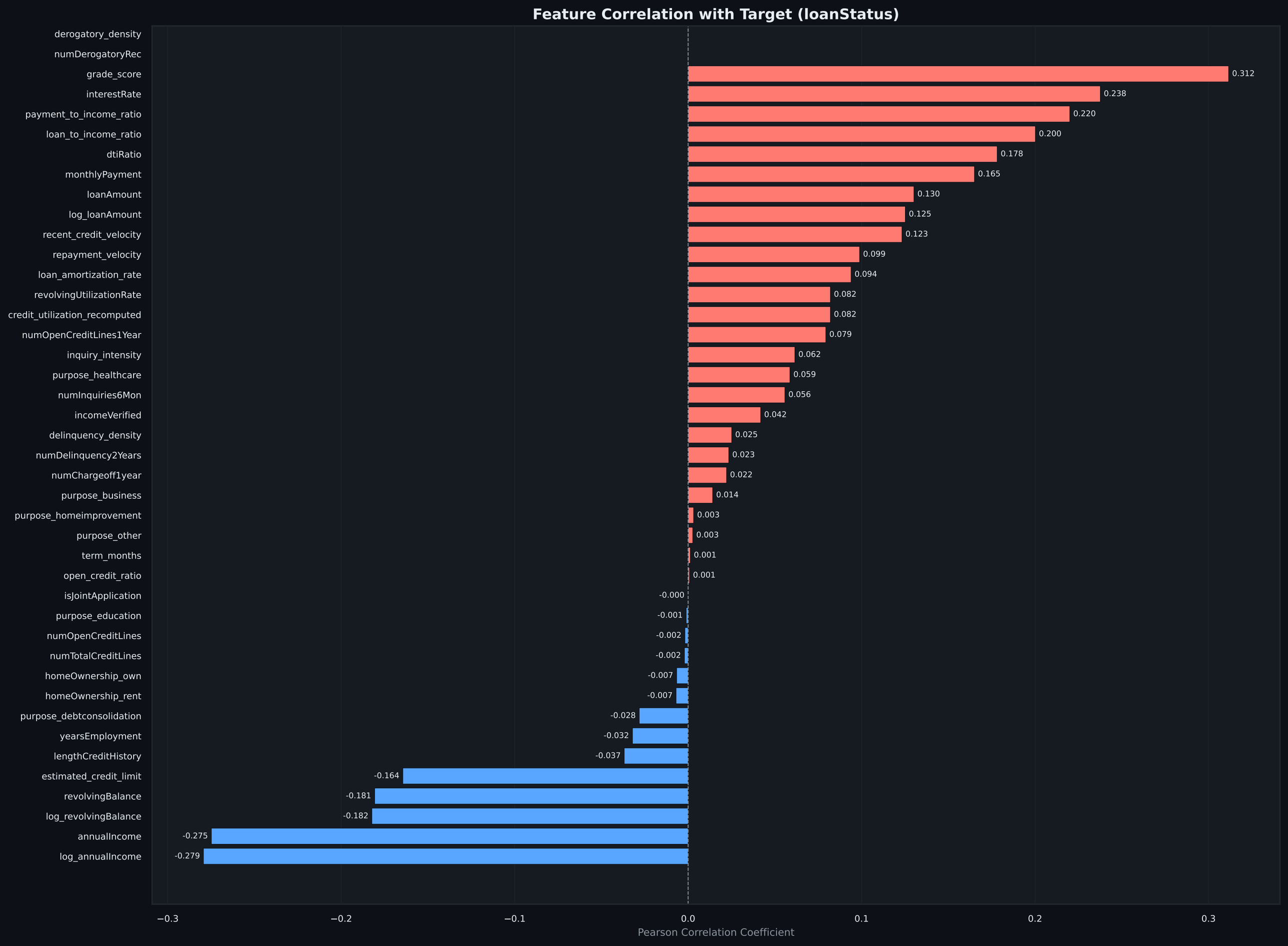
Feature Distributions (Page 4/4)



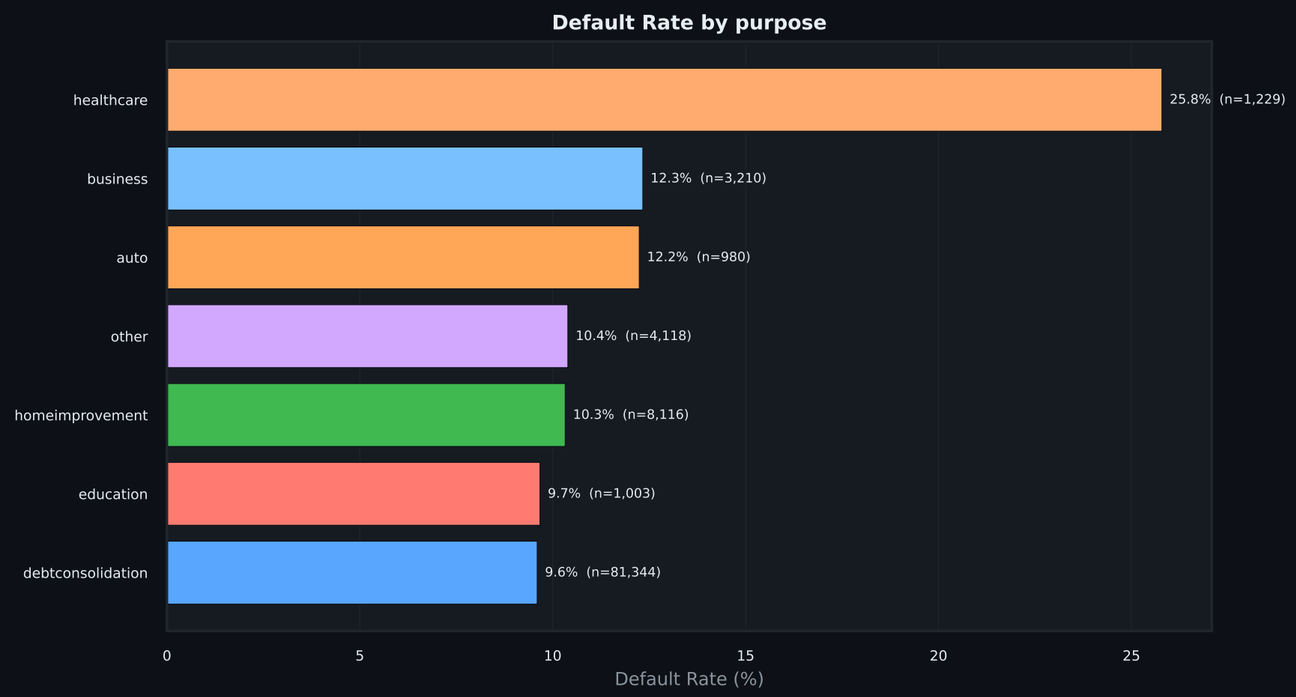
Key Feature Distributions vs Target (Box-plots)

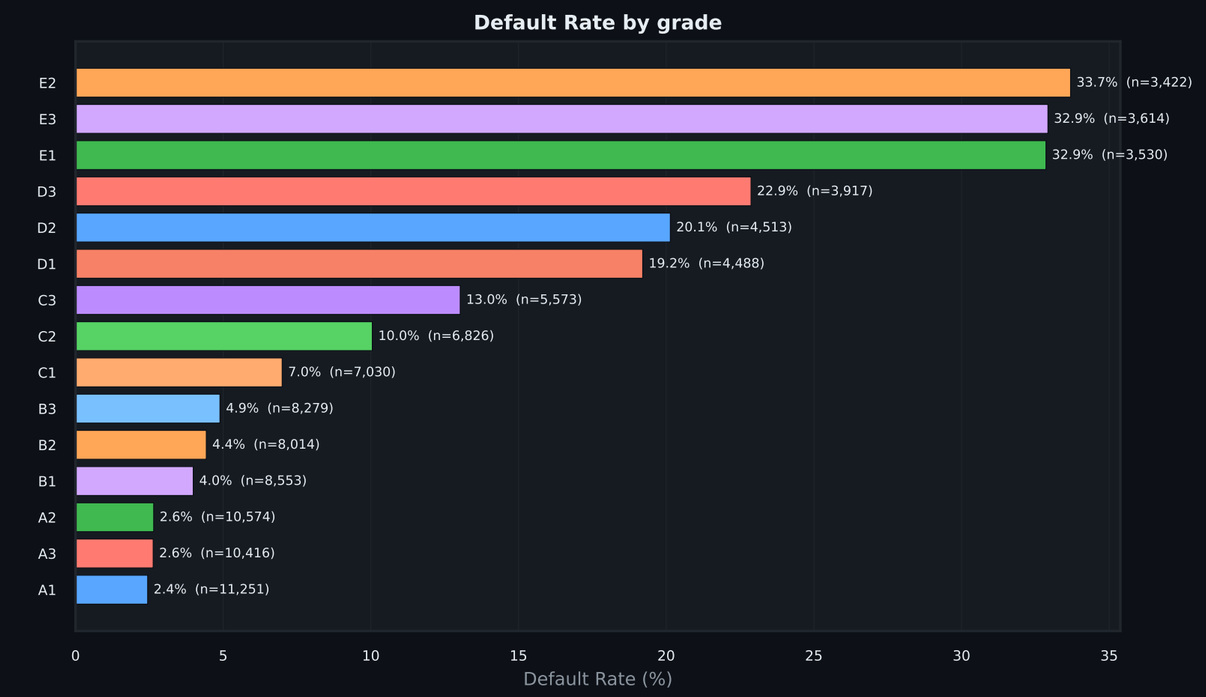
Correlation Heatmap – Pairwise Pearson Correlation

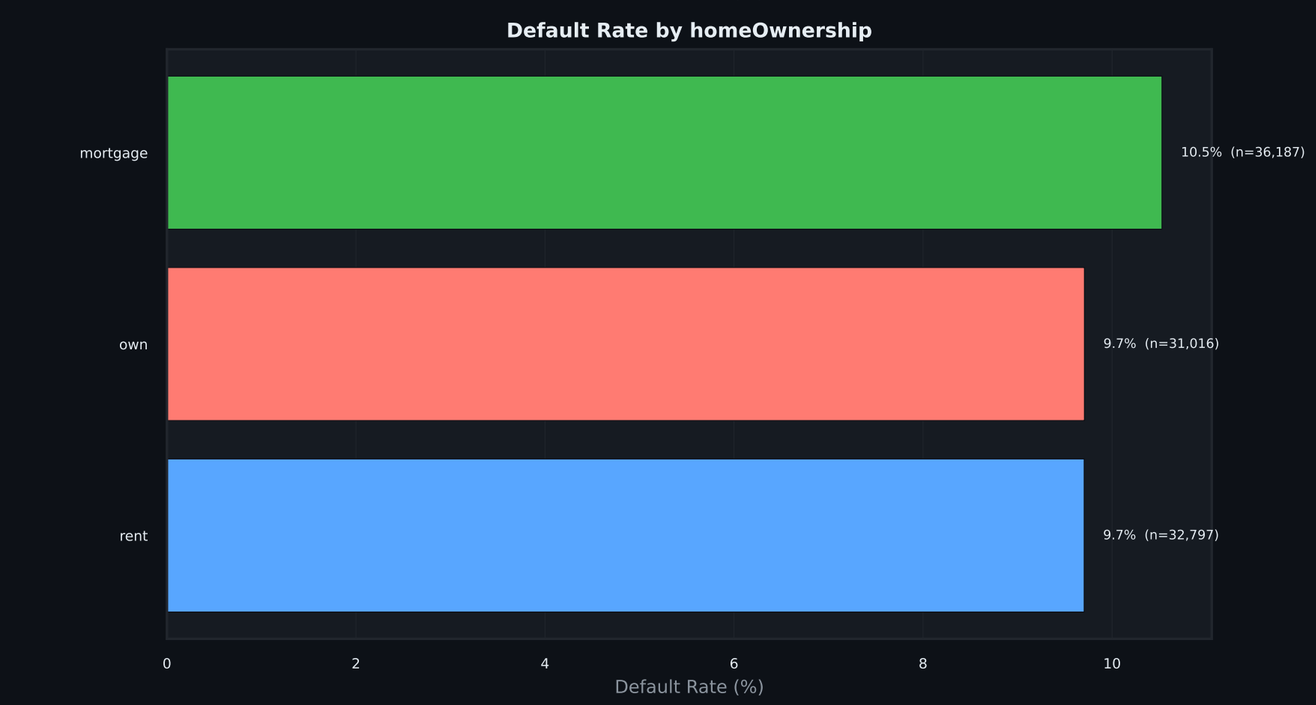



Feature Correlation with Target (loanStatus)
derogatory_density
numDerogatoryRec
grade_score 0.312
interestRate 0.238
payment_to_income_ratio 0.220
loan_to_income_ratio 0.200
dtiRatio 0.178
monthlyPayment 0.165
loanAmount 0.130
log_loanAmount 0.125
recent_credit_velocity 0.123
repayment_velocity 0.099
loan_amortization_rate 0.094
revolvingUtilizationRate 0.082
credit_utilization_recomputed 0.082
numOpenCreditLines1Year 0.079
inquiry_intensity 0.062
purpose_healthcare 0.059
numInquiries6Mon 0.056
incomeVerified 0.042
delinquency_density 0.025
numDelinquency2Years 0.023
numChargeoff1year 0.022
purpose_business 0.014
purpose_homeimprovement 0.003
purpose_other 0.003
term_months 0.001
open_credit_ratio 0.001
isJointApplication -0.000
purpose_education -0.001
numOpenCreditLines -0.002
numTotalCreditLines -0.002
homeOwnership_own -0.007
homeOwnership_rent -0.007
purpose_debtconsolidation -0.028
yearsEmployment -0.032
lengthCreditHistory -0.037
estimated_credit_limit -0.164
revolvingBalance -0.181
log_revolvingBalance -0.182
annualIncome -0.275
log_annualIncome -0.279
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Pearson Correlation Coefficient

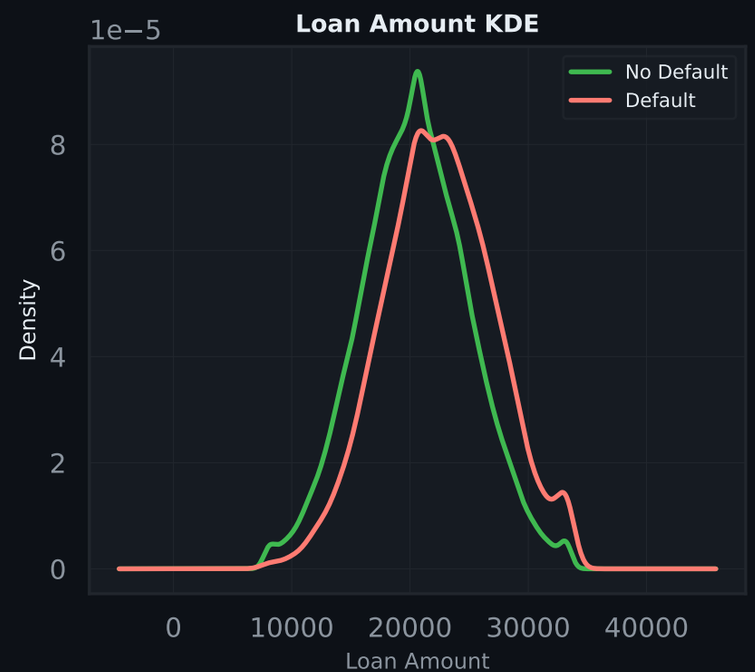
Categorical Features vs Default Rate

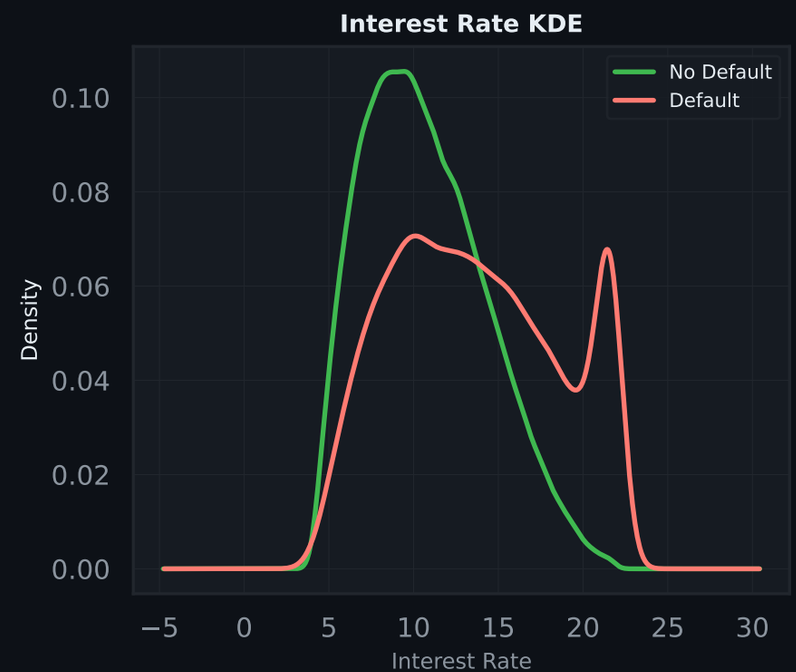


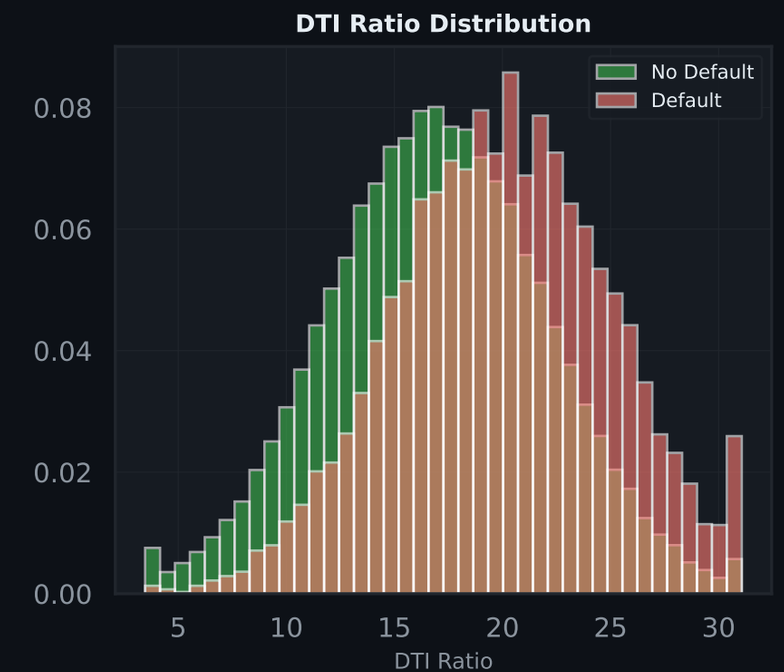


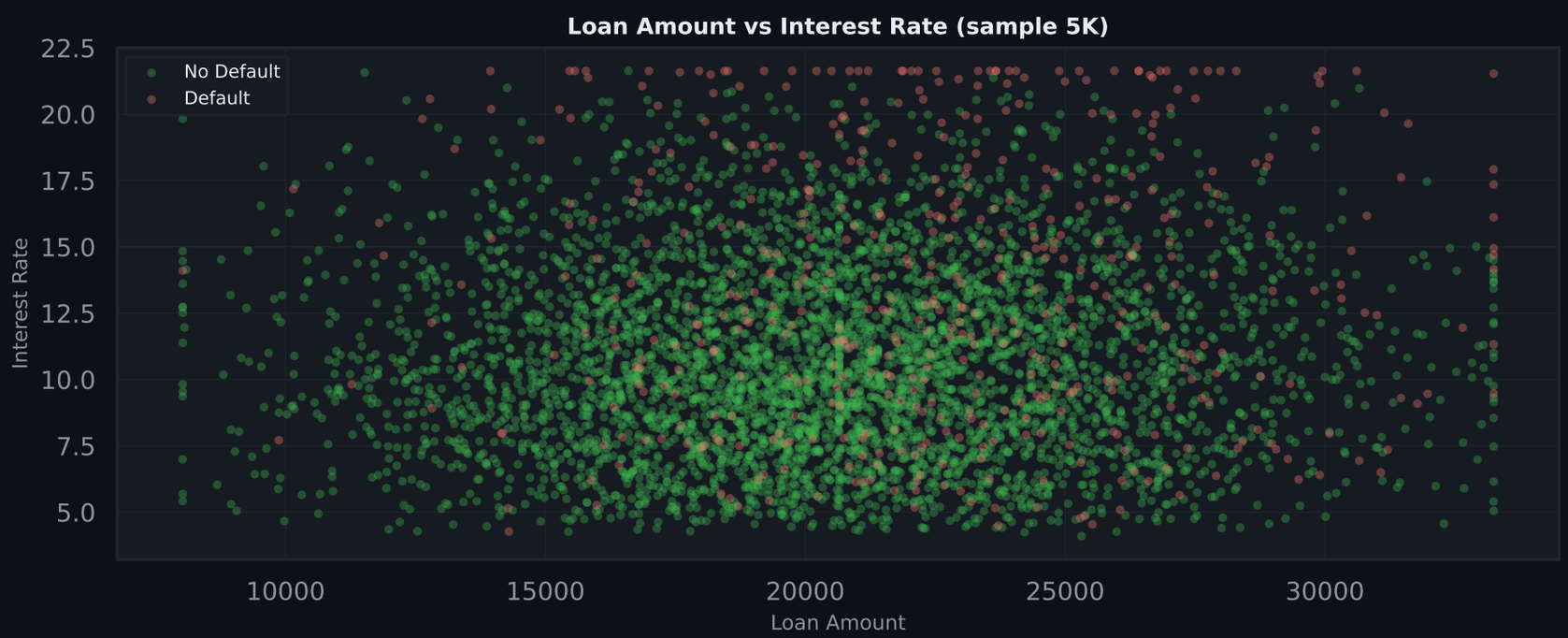


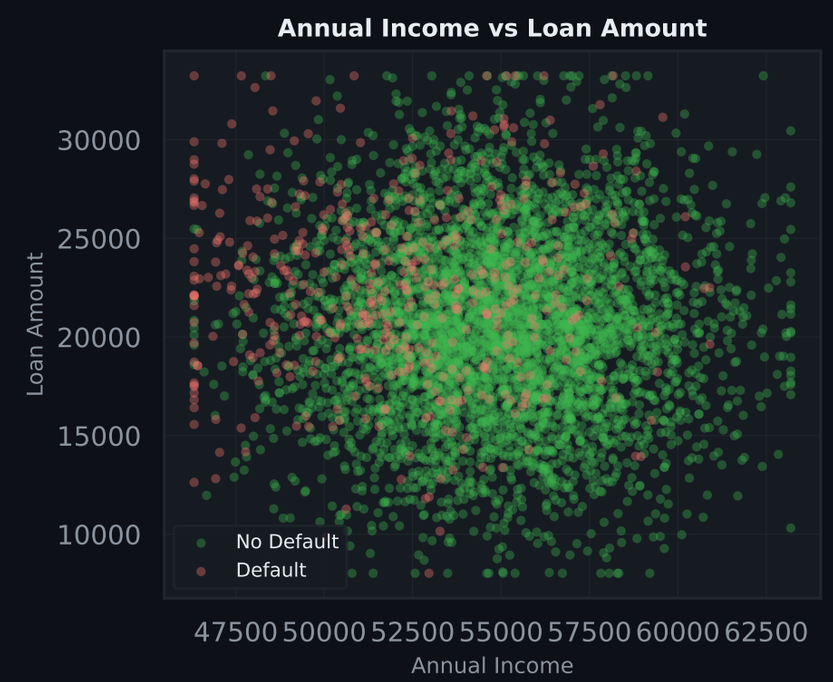
Interest Rate & Loan Amount Deep-Dive



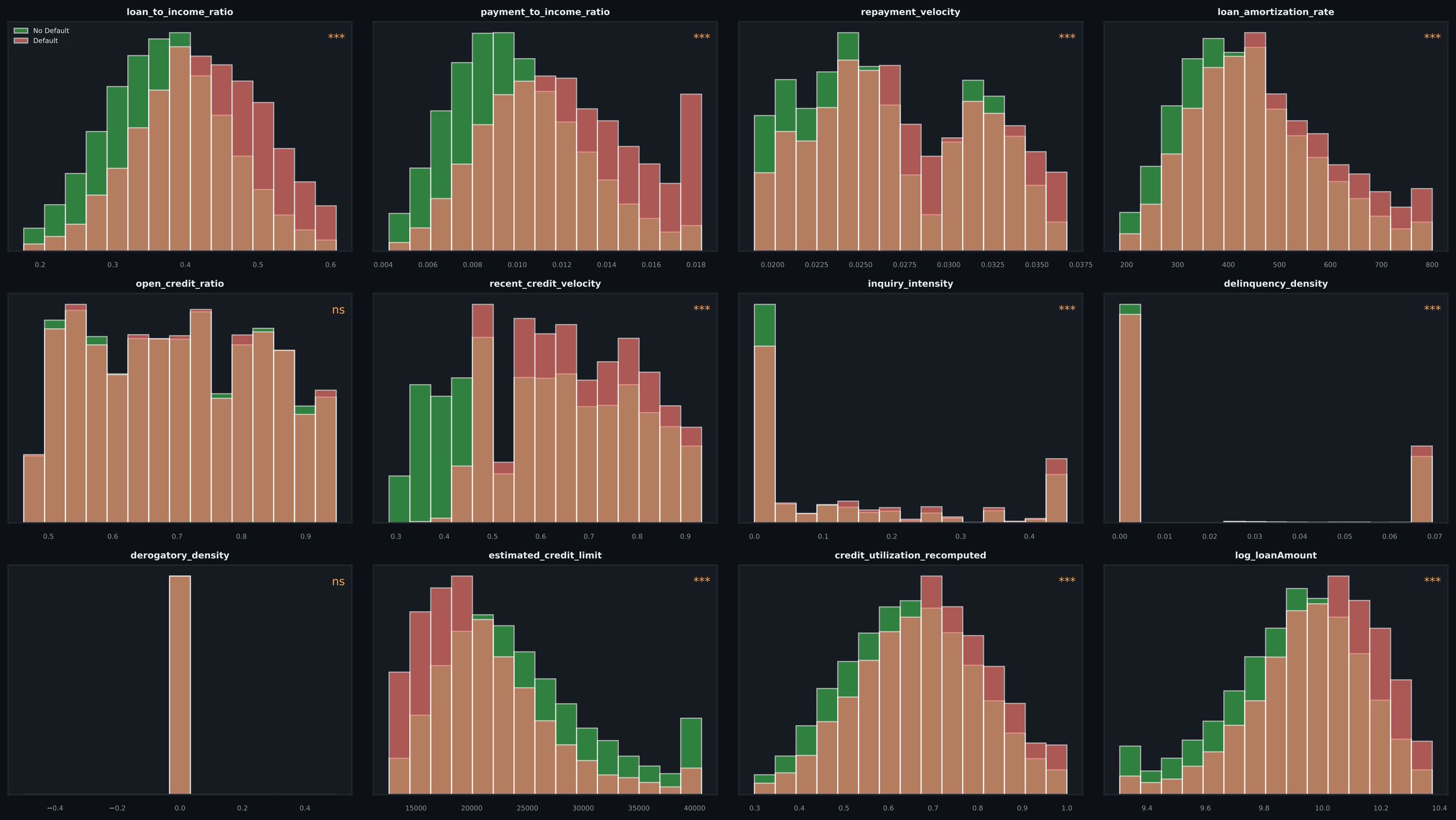




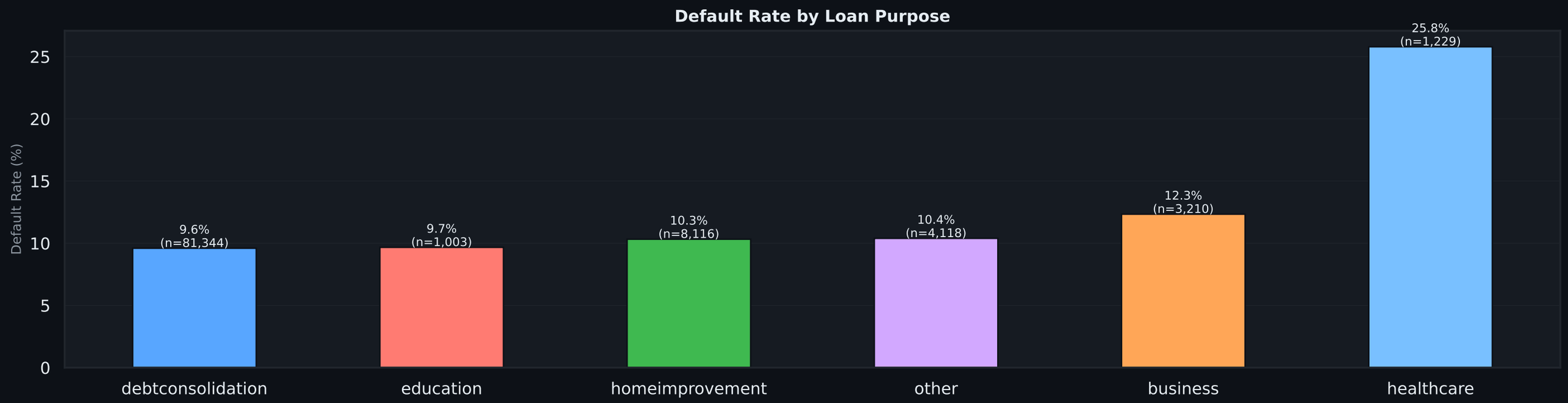


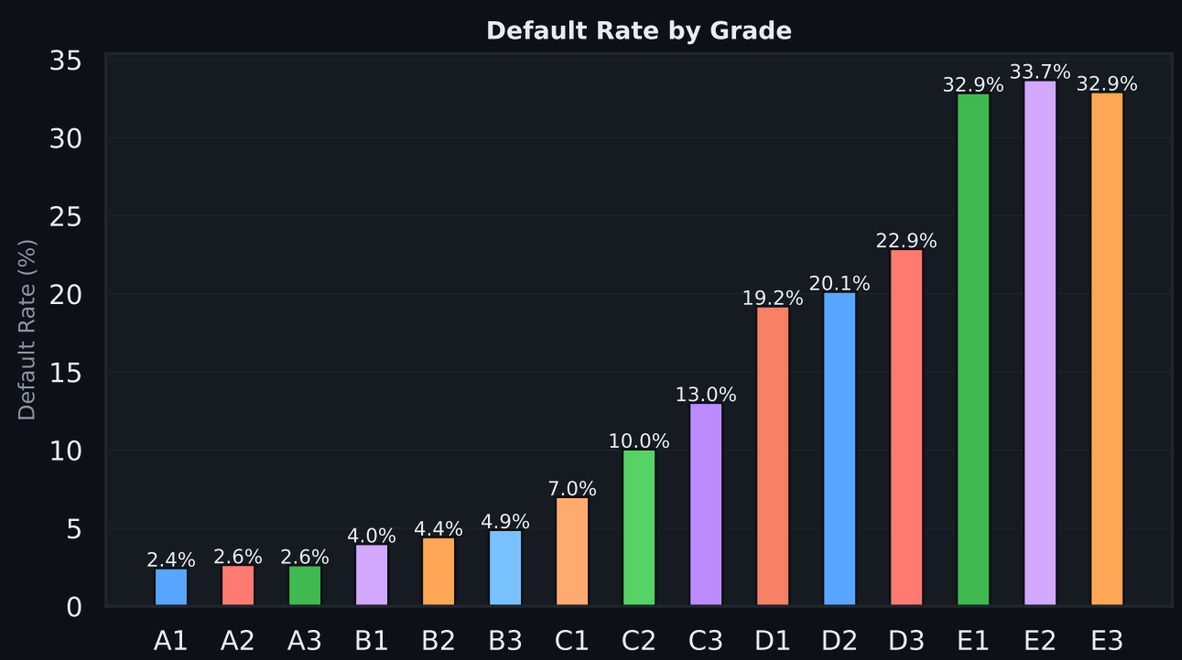


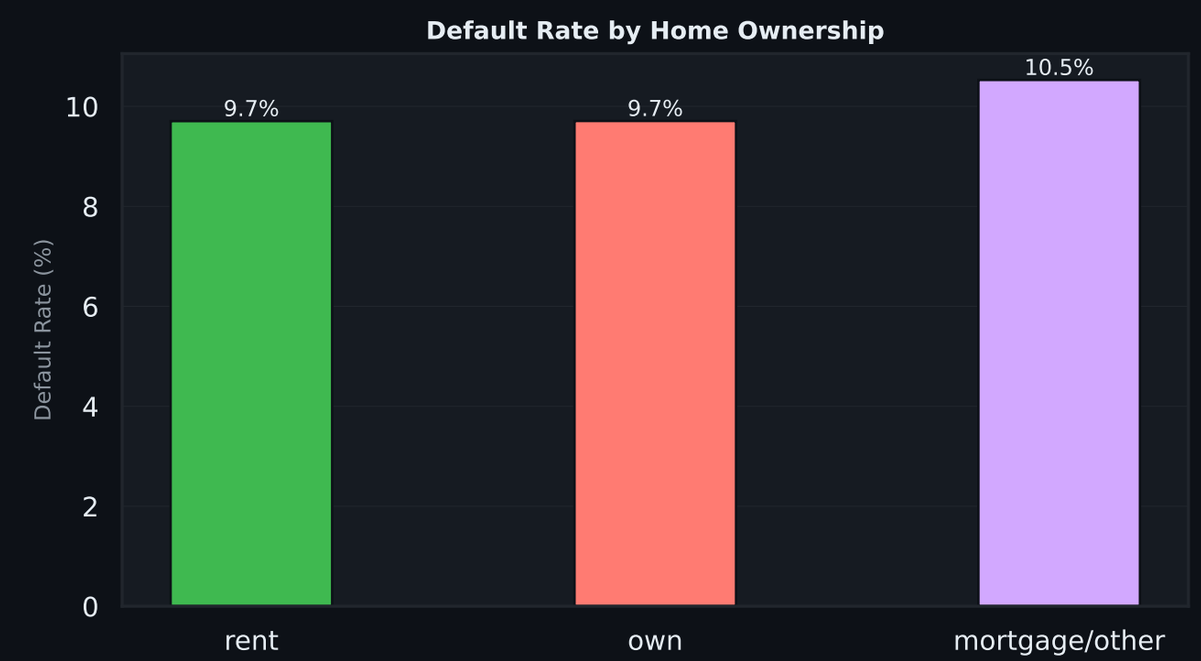
Engineered Features vs Target



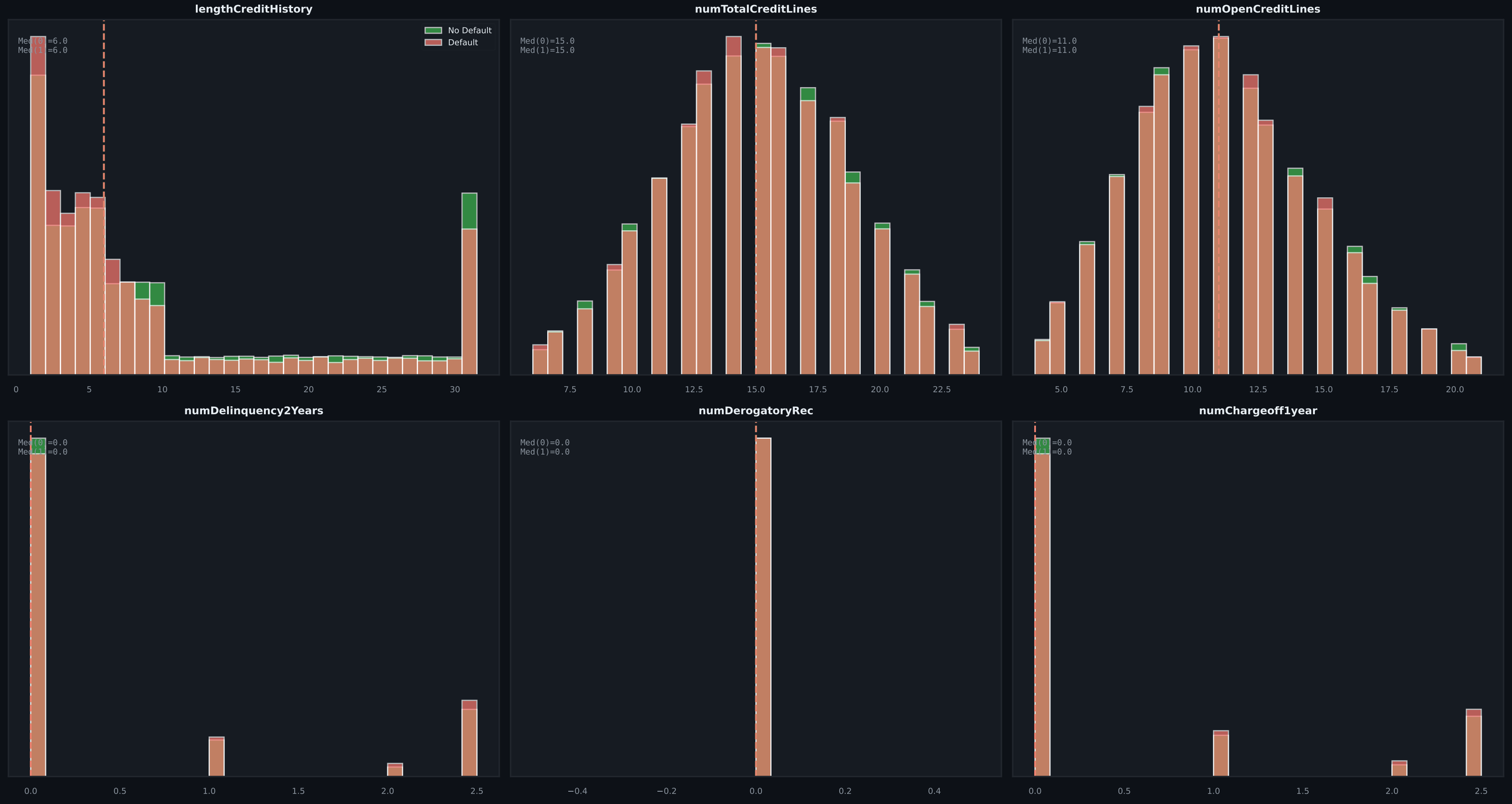
Purpose & Grade Analysis







Credit History & Delinquency Analysis



EDA Summary – Key Findings

10.00%
Overall Default Rate

13.67%
Avg Interest Rate (Default)

10.65%
Avg Interest Rate (No Default)

19.99
Avg DTI Ratio (Default)

16.98
Avg DTI Ratio (No Default)

100,000
Total Records

Top Positive Correlations with Default:

- grade_score: +0.312
- interestRate: +0.238
- payment_to_income_ratio: +0.220
- loan_to_income_ratio: +0.200
- dtiRatio: +0.178

Top Negative Correlations with Default:

- log_annualIncome: -0.279
- annualIncome: -0.275
- log_revolvingBalance: -0.182
- revolvingBalance: -0.181
- estimated_credit_limit: -0.164